Effect of the catalyst support for Pt and Ir based catalysts on the NO_x formation during combustion of NH_3 SOFC off-gas

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Introduction

Ammonia has attracted a lot of interest as a potential green and renewable hydrogen carrier or energy vector. It can be cracked to provide hydrogen and it is also possible to use it directly for power generation in high-temperature solid oxide fuel cells (SOFCs).[1-3]

In this study, the effect of different catalyst supports on the performance of iridium and platinum-based catalysts for the off-gas combustion of a NH_3 SOFC with focus on NO_x formation was investigated.

Results

The supported catalyst powders were prepared by impregnating the catalyst support with aqueous solutions of $(NH_3)_4Pt(NO_3)_2$ or $IrCl_3*xH_2O$. After calcination at 450°C, the catalyst powders were deposited into the microchannels as washcoats using an aqueous suspension. The plates were calcined at 450°C and tested for the combustion of NH_3 SOFC off-gas at different temperatures (100 - 800°C).

Table 1: Catalyst characterization data.

Catalyst	Content (target) wt.%	Content (measured via XRF) wt.%	BET surface area m^2g^{-1}	Particle Size (TEM) nm
Pt/Al2O3	5	4.82 ± 0.14	150	2.35
Pt/SiO2	5	4.68 ± 0.14	90	2.37
Pt/TiO2	5	4.81 ± 0.14	36	1.36
Ir/Al2O3	5	4.98 ± 0.15	152	>6
Ir/SiO2	5	4.80 ± 0.14	87	3.72
Ir/TiO2	5	5.37 ± 0.16	37	1.60

Regardless of the catalyst support, the platinum-based catalysts showed a complete conversion of H_2 and NH_3 above 150°C, whereas this could only be observed above 400°C for the iridium-based catalysts (see Fig. 1). The catalyst supports showed a strong influence on NO_x formation. For all catalysts tested, the highest NO_x formation was observed for the Al_2O_3 support, followed by SiO_2 and TiO_2 (see Fig. 2 & 3).

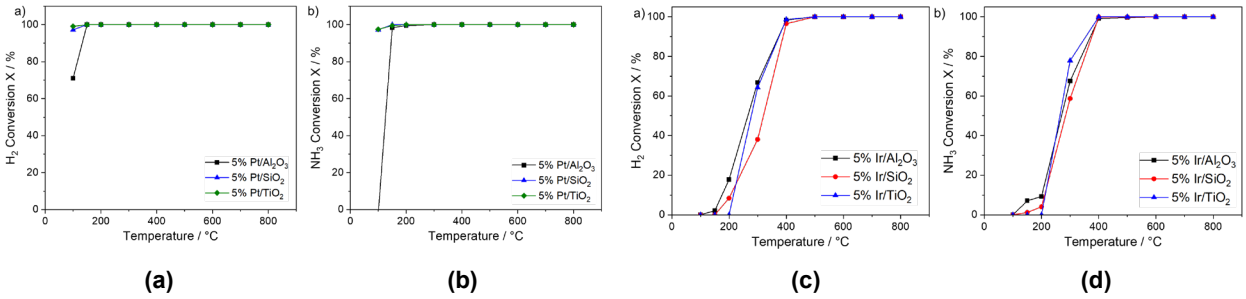

Figure 1: Hydrogen and ammonia conversion vs. reaction temperature for 5% platinum (a, b) and 5% iridium (c, d) on different supports, WHSV = 600 L/g h.

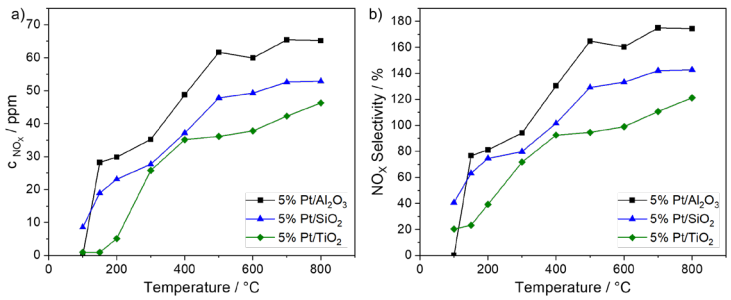

Figure 2: NO_x concentration and NO_x selectivity vs. reaction temperature for 5% platinum on different supports, WHSV = 600 L/g*h.

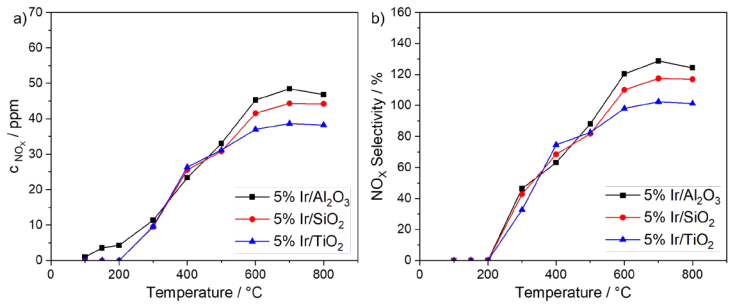

Figure 3: NO_x concentration and NO_x selectivity vs. reaction temperature for 5% iridium on different supports, WHSV = 600 L/g*h.

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